

1. Перевести из 10 в 16 систему 12345678, 1000000

$$\begin{array}{r}
 12345678 \mid 16 \\
 - 12345664 \mid 771604 \mid 16 \\
 \hline
 14(E) \mid 771600 \mid 48225 \mid 16 \\
 \mid 4 \mid 48224 \mid 3014 \mid 16 \\
 \mid 1 \mid 3008 \mid 188 \mid 16 \\
 \mid 6 \mid 126 \mid 11(B) \\
 \mid 12(C)
 \end{array}$$

$$12345678_{10} = BC614E_{16}$$

$$\begin{array}{r}
 1000000 \mid 16 \\
 - 1000000 \mid 62500 \mid 16 \\
 \hline
 0 \mid 62496 \mid 3906 \mid 16 \\
 \mid 4 \mid 3904 \mid 244 \mid 16 \\
 \mid 2 \mid 240 \mid 15(F) \\
 \mid 4
 \end{array}$$

$$1000000_{10} = F4240_{16}$$

2. Перевести из 16 в 10 систему 12345678, 1000000

$$\begin{array}{cccccccc} 7 & 6 & 5 & 4 & 3 & 2 & 1 & 0 \\ 1 & 2 & 3 & 4 & 5 & 6 & 7 & 8 \end{array}_{16} = 305419896_{10}$$

$$1 \cdot 16^7 + 2 \cdot 16^6 + 3 \cdot 16^5 + 4 \cdot 16^4 + 5 \cdot 16^3 + 6 \cdot 16^2 + 7 \cdot 16^1 + 8 \cdot 16^0 = 305419896$$

$$\begin{array}{cccccc} 6 & 5 & 4 & 3 & 2 & 1 & 0 \\ 1 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}_{16} = 16777216_{10}$$

$$1 \cdot 16^6 + 0 \cdot 16^5 + 0 \cdot 16^4 + \dots + 0 \cdot 16^0 = 16777216$$

3. Записать в виде логического выражение ответ Винни Пуха:

“Сгущенного молока и меда и можно без хлеба”

A - молоко

B - мед

C - хлеб

A	B	C	!C	$A \vee B \vee !C$
0	0	0	1	0
0	0	1	0	0
0	1	0	1	0
0	1	1	0	0
1	0	0	1	0
1	0	1	0	0
1	1	0	1	1
1	1	1	0	0

5. Прислать скриншот игры Игра на Яндекс или Boolean games by July Sudarenko нужен VPN или

The diagram illustrates the implementation of the XOR operation using a truth table and a logic expression. The logic expression is $X = IB + A*B + !A*!B$. The truth table below shows the values of A, B, IB, A*B, !A*!B, and X for all possible combinations of A and B.

A	B	IB	A*B	!A*!B	X
0	0	1	0	1	1
0	1	0	0	0	0
1	0	1	0	0	1
1	1	0	1	0	1

The screenshot shows a game interface for converting binary to decimal. On the left, a 'BIN' box contains the binary sequence '01100000', with a '0' box highlighted. A large black arrow points from the 'BIN' box to a 'HEX' box. In the center, a 2x2 grid of buttons shows the decimal value '60'. Above this grid, a green-bordered box displays 'DEC 96'. On the right, a '!' box is visible. Navigation icons (back, forward, home, and a large 'X') are on the left, and a play button and a circular arrow icon are on the right.

The screenshot shows a game interface with a dark blue background. On the left, there are two large, stylized icons: a grey 'X' and a green checkmark. In the center, there is a hex editor window with a green border and a dark blue background. The text 'HEX BD' is displayed in the center of the hex editor. Below the hex editor, there is a grid of 16 buttons arranged in two rows of four. The top row of buttons contains black upward-pointing triangles. The bottom row of buttons contains black downward-pointing triangles. In the center of the grid, there are four buttons with the following text: '-', '0', '6', and '7'. To the right of the grid, there is a large, stylized grey 'G' icon and a large, stylized grey exclamation mark. At the bottom of the screen, there is a row of buttons. The first button is labeled 'BIN' and has a large black downward-pointing arrow above it. The other buttons in the row contain the following text: '1', '0', '1', '1', '1', '1', '0', and '1'.